

Vishal Singh Yadav

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SUMMARY

Senior AI/ML Engineer with 5+ years of experience building and delivering production ML and GenAI systems end-to-end. Strong in Python backend development for AI platforms (APIs, microservices, data pipelines, distributed processing) with cloud-native MLOps/DevOps (Docker, Kubernetes, CI/CD) and observability (logging, metrics, tracing).

Built GenAI systems (RAG, NLQ-to-SQL, agents) with a focus on reliability, latency, and stakeholder-facing delivery.

EXPERIENCE

- **Tredence Inc.** Hyderabad, India
Sep 2025 – Present
Associate Manager (Senior Data Scientist)
 - Ran a detailed latency and architecture review of an enterprise conversational-analytics LLM assistant (LangGraph/LangChain orchestration over Azure OpenAI and Databricks), and traced its roughly 158s median response time, well above the 80s target, to 11 serial LLM calls and unbounded retry loops.
 - Proposed a deterministic, AST-based query pipeline (unified intent/slot extraction, term and metric resolution, and validated AST-to-SQL generation) that reduced the path from 11 LLM calls to 3 and projected roughly an 80% latency improvement; shared the assessment and a delivery plan with cost and resourcing estimates with engineering leadership, who aligned on the direction.
 - Built an LLM-based classification pipeline to organize 1.8M+ purchase-order line items into a 110-category cost taxonomy, using hash-based deduplication (from 1.8M down to about 111K distinct cases) and a multi-threaded, retry-driven design to keep it reliable and fully covered at that scale.
 - Developed a delay-prediction approach for a large-scale construction analytics platform that blended a percentile baseline with a DAG critical-path estimator; it improved mean absolute error by about 30% (from 127 to 89 days) and surfaced 80% more at-risk projects than the previous method, across 6,000+ active projects.
 - Built GenAI Accelerator (Internal) deliverables for internal evaluation, NLQ-to-SQL with guardrails, RAG-based document Q&A, and tool-using agent flows with bounded retries and structured outputs.
 - Analyzed regional cost variation across capital markets, using statistical outlier detection and vendor-concentration analysis to show that exclusive-vendor pricing, not schedule or scope creep, drove most of the highest-cost ($2-3\sigma$) sites.
- **Carelon Global Solutions (Elevance Health)** Hyderabad, India
Jul 2023 – Sep 2025
Associate Data Scientist III
 - Built forecasting models for the Cost of Care platform across 60M+ members, delivering 3-month state-level utilization forecasts with Bayesian uncertainty quantification.
 - Reduced forecast error variance by ~12% through model and feature improvements, improving stability of downstream planning decisions.
 - Developed NLP- and probabilistic ML solutions across clinical, financial, and demographic data to improve underwriting accuracy and decision quality.
 - Reduced underwriting turnaround time by 20% and increased automated coverage from 48% to 81% by improving data extraction and risk modeling workflows.
 - Enabled ~\$1.8M in projected annual savings through model-driven process optimization and higher automation coverage.
 - Built a secure clinical Q&A assistant using LLAMA2, LangChain, and QLoRA-tuned RAG; improved runtime performance by ~31% via optimization across retrieval/inference steps.
 - Implemented NLQ-to-SQL pipelines for analytics workflows, achieving ~95% query accuracy with schema-aware constraints and validation.
 - Deployed and operated production ML systems across AWS, Azure, and on-prem environments using Dockerized services, CI/CD pipelines, and automated validation.
 - Implemented monitoring and retraining workflows to support production readiness, reliability, and continuous model performance tracking.
- **Qulabs Software India Pvt Ltd** Hyderabad, India
Mar 2023 – Jul 2023
Machine Learning Engineer
 - Built a semantic search and document QA system using BERT embeddings and vector search over 100K+ documents, enabling fast, context-aware retrieval without reliance on LLMs.
 - Deployed NLP models for production use cases and mentored junior engineers on ML workflows, code quality, and deployment practices, improving maintainability and reuse across projects.
- **Krama Lab, IIT Hyderabad (Machine Learning Research)** Hyderabad, India
Jan 2020 – Dec 2022
Research Assistant
 - Conducted applied research on graph neural networks, point cloud segmentation, and few-shot learning, developing model architectures for structured reasoning and sparse-data industrial settings.

- Designed and evaluated GNN-based approaches for knowledge graph completion and link prediction under low-resource constraints, focusing on representation learning and generalization.

SELECTED CONSULTING EXPERIENCE

- **Independent Consulting (Part-time)** Remote / India
Consultant, Machine Learning & GenAI Systems *Jul 2025 – Sep 2025*
 - Delivered components for a client-facing agent-based analytics system, integrating natural-language interfaces with existing data, sensor, and optimization infrastructure.
 - Implemented schema-aware query routing and tool-orchestrated workflows for real-time data access, with guardrails to improve reliability in production-like environments.
 - Performed system-level analysis to address latency, noisy data, and operational constraints; translated findings into engineering recommendations for production readiness.

EDUCATION

- **Indian Institute of Technology Hyderabad** Hyderabad, India
M.Tech (by Research) in Computer Science and Engineering GPA: 8.38 / 10 *Jan 2020 – Dec 2022*
- **Guru Gobind Singh Indraprastha University** New Delhi, India
B.Tech in Computer Science and Engineering Percentage: 71.2% *Jul 2014 – May 2018*

SKILLS

- **AI Systems & GenAI:** LLMs, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings & Vector Search (FAISS), NLQ-to-SQL, AI Agents, Tool/Function Calling, Guardrails & Validation, Fine-tuning (LoRA, QLoRA)
- **Machine Learning & NLP:** Transformers, BERT, spaCy, Semantic Search, Time-Series Forecasting, Anomaly Detection, Graph Neural Networks, Knowledge Graphs, Few-Shot Learning, PyTorch, TensorFlow, Scikit-learn, XGBoost
- **Backend & APIs:** Python, SQL, FastAPI, REST APIs, Microservices, Distributed services for data/ML workloads, Model Serving, ONNX
- **MLOps / LLMOps / DevOps:** Docker, Kubernetes, CI/CD, Monitoring, Observability (logging/metrics/tracing), Automated validation, Performance tuning
- **Data Engineering & Platforms:** PySpark, Snowflake, Distributed Computing, ETL Pipelines, Time-Series Analysis
- **Programming Languages:** Python, SQL, C++, Bash

PROJECTS

- **CodeChakra: AI-Powered PR Review Assistant (Open-source):** Built an AI-assisted PR review system that combines static code analysis (quality, security signals, performance/latency checks) with LLM-based reasoning to generate actionable engineering feedback. Designed a modular agent workflow with secure I/O boundaries and CI integration, enabling local/self-hosted deployment via Docker for high-trust environments. *(In progress)*

ACHIEVEMENTS

- **Go Above Award (x2) – Carelon Global Solutions:** Recognized twice for high-impact delivery under tight timelines, including rapid launch and production deployment of business-critical analytics products.
- **Leadership Recognition – Carelon Global Solutions:** Acknowledged by senior leadership for fast-tracked execution and delivery ownership on high-visibility initiatives supporting decision making.
- **Mentor – AI and Advanced Technologies:** Mentored working professionals via IIT Hyderabad & NSE Talentsprint program on ML workflows and production-grade AI system development.

CERTIFICATIONS

- **GenAI / LLM Engineering – Tredence / UNext:** Completed an internal GenAI engineering program focused on building production-oriented LLM applications: RAG, embeddings/vector search, NLQ-to-SQL, tool-using agents, guardrails/validation, evaluation signals, and deployment considerations (latency, reliability, monitoring).
- **Advanced Certification in Generative AI – Carelon / Prizmato:** Production deployment of LLMs, QLoRA fine-tuning, RAG system design, and compliance-aware GenAI workflows.
- **NVIDIA Deep Learning Institute:** NLP and Deep Learning (Transformers), AI for Anomaly Detection and Data Engineering, Accelerated Computing and Data Science.

PUBLICATIONS

- **Context-Aware Question Routing in Community Question Answering Sites:** *Vishal Singh Yadav, Manish Singh* May 2023. Proposed a deep learning-based approach for intelligent question routing using contextual embeddings and semantic similarity to improve expert matching in community Q&A platforms.